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ubuntu@ip-172-31-10-237:~$ #Lab 54: Using df and du
ubuntu@ip-172-31-10-237:~$ #Objectives:Understand and utilize the df command to check disk space usage.Learn how to use the du command to assess directory usage. Analyze the results provided by each command to manage disk space effectively.
ubuntu@ip-172-31-10-237:~$ #Overview:Efficient disk space management is crucial for maintaining system performance. Two fundamental commands used for this purpose are df (disk free) and du (disk usage). This lab will guide you through practical applications of these commands to monitor and manage disk space on a Linux system.
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ #Task1
ubuntu@ip-172-31-10-237:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root         29G  7.1G   21G  26% /
tmpfs            1.9G  8.0K  1.9G   1% /dev/shm
tmpfs            768M  1.3M  767M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.1K  119K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M  240M  580M  30% /boot
/dev/nvme0n1p15  105M  6.2M   99M   6% /boot/efi
tmpfs            384M  128K  384M   1% /run/user/1000
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ #Purpose:The df command displays the amount of disk space available on the file system.
The -h option stands for human-readable format, which presents the data in gigabytes (G), megabytes (M), etc.
-bash: syntax error near unexpected token `('
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ ^C
ubuntu@ip-172-31-10-237:~$ #ANALYSIS OF df -h: Your system is in good health regarding disk space.                                    Total root filesystem usage is moderate at 26%.
Critical partitions (especially /boot and /boot/efi) have comfortable free space.
No immediate risk of running out of space.
The system is using an NVMe SSD (/dev/nvme0n1), which is excellent for performance.
Total: command not found
-bash: syntax error near unexpected token `('
No: command not found
-bash: syntax error near unexpected token `('
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ #task2
ubuntu@ip-172-31-10-237:~$ # Checking Directory Usage with du
ubuntu@ip-172-31-10-237:~$ du -sh /home/user/Documents
du: cannot access '/home/user/Documents': No such file or directory
ubuntu@ip-172-31-10-237:~$ cd /
ubuntu@ip-172-31-10-237:/$ ls
bin                boot  etc   lib                lib64       media  opt   root  sbin
snap  sys  usr
bin.usr-is-merged  dev   home  lib.usr-is-merged  lost+found  mnt    proc  run   sbin.usr-is-
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merged   srv    tmp   var
ubuntu@ip-172-31-10-237:/$ cd /home/user
-bash: cd: /home/user: No such file or directory
ubuntu@ip-172-31-10-237:/$ cd /home
ubuntu@ip-172-31-10-237:/home$ ls
ubuntu
ubuntu@ip-172-31-10-237:/home$ cd ubuntu
ubuntu@ip-172-31-10-237:~$ ls
Desktop     Downloads   NICE-GPG-KEY   Public      Videos                                   snap
Documents   Music       Pictures       Templates   nice-dcv-2025.0-20103-ubuntu2404-x86_64
ubuntu@ip-172-31-10-237:~$ cd ..
ubuntu@ip-172-31-10-237:/home$ cd ~
ubuntu@ip-172-31-10-237:~$ #Task2
ubuntu@ip-172-31-10-237:~$ du -sh /home/ubuntu/Documents
4.0K    /home/ubuntu/Documents
ubuntu@ip-172-31-10-237:~$ #Purpose:The du command estimates the file space usage.The -s flag provides a summary (just the total size# ).The -h option formats the output in a human-readable manner.
ubuntu@ip-172-31-10-237:~$ #ANALYSIS OF df -sh: The root filesystem (/) is only 26% used (7.1 GB out of 29 GB).A 4 KB Documents folder is negligible — it represents ~0.00005% of your total used space
ubuntu@ip-172-31-10-237:~$ #subtask2: Identifying large files or folders
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ du -ah /home/ubuntu/Documents | sort -rh | head -n 10
4.0K    /home/ubuntu/Documents
ubuntu@ip-172-31-10-237:~$ df -ah /home/ubuntu
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G  7.1G   21G  26% /
ubuntu@ip-172-31-10-237:~$ cron jobs
cron: can't open or create /var/run/crond.pid: Permission denied
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ #CONCLUSION:By mastering the df and du commands, I can efficiently monitor and manage disk usage on your Linux system. Regular use of these commands will help you prevent disk space issues and maintain optimal system performance. Implement strategies to archive or delete unused files based on the insights gained from these analyses.
ubuntu@ip-172-31-10-237:~$
ubuntu@ip-172-31-10-237:~$ #LabCompleted
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